

PET and Gestational Hypertension 03: Diagnosis Without Chaos

Companion reading handout for simultaneous listening.

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Opening Case

Let us begin in triage.

A patient is thirty four weeks. The automated cuff reads one hundred fifty two over ninety eight. She says she has a headache, but she also has not eaten and did not sleep. The urine dipstick is one plus. The fetus is moving. The room is busy.

This is where hypertensive disease becomes chaotic if you do not have a diagnostic order.

The order is simple.

First, confirm the blood pressure correctly. Second, decide whether there are severe features. Third, quantify proteinuria properly. Fourth, assess maternal organs. Fifth, assess the fetus and placenta. Sixth, use biomarkers only as a clinical adjunct, not as a substitute for judgment.

Start with blood pressure technique.

The patient should be seated, with feet supported and the arm at heart level. The cuff size matters. If the cuff is too small, the pressure may be falsely high. If the patient is lying flat late in pregnancy, supine hypotension and positioning artifacts can confuse the reading. ISSHP recommends using devices validated in pregnancy and pre-eclampsia, and measuring both arms initially.

If the pressure is non-severe, persistence matters. ACOG and ISSHP use repeated readings, typically at least four hours apart, to confirm true hypertension. That protects the patient from being mislabeled because of pain, anxiety, movement, or measurement error.

But if the pressure is severe, do not wait four hours.

Severe means systolic at least one hundred sixty, diastolic at least one hundred ten, or both. ISSHP says repeat within fifteen minutes. ACOG says confirm within minutes when needed so treatment is not delayed. The reason is stroke prevention. A four-

hour confirmation rule makes sense for diagnosis. It is dangerous for severe hypertension.

Once hypertension is real, ask whether this is isolated hypertension or pre-eclampsia.

Proteinuria is one route into the diagnosis. The classic threshold is at least three hundred milligrams in a twenty four hour urine collection. But the modern workflow usually does not need to wait twenty four hours. ACOG accepts a spot protein to creatinine ratio of at least zero point three. NICE, ISSHP, and SOMANZ use protein to creatinine ratio at least thirty milligrams per millimole, or albumin to creatinine ratio at least eight milligrams per millimole.

The point is not that these are magical numbers. They are practical thresholds that identify clinically meaningful proteinuria quickly enough to guide care.

Dipstick is not enough. It is cheap and available, but false positives and false negatives are common. ACOG warns about poor reliability. NICE says that if dipstick is positive, quantify it. SOMANZ says dipstick can screen, but cannot confidently diagnose.

Now the most important proteinuria rule.

Proteinuria diagnoses. It does not stage severity.

Once pre-eclampsia is established, repeating the amount of protein usually does not decide delivery. The patient with three grams is not automatically more urgent than the patient with one gram. Severity is driven by blood pressure, platelets, creatinine, liver injury, pulmonary edema, neurologic symptoms, fetal status, and gestational age.

Next, check maternal organs.

The minimum lab set is platelets, creatinine, and liver enzymes. If there is concern for HELLP syndrome, DIC, or hemolysis, add LDH, blood smear, coagulation studies, and fibrinogen according to local practice.

Platelets below one hundred thousand are a severe feature in ACOG and the Hadassah protocol. Creatinine above one point one milligrams per deciliter, or doubled from baseline, is a severe feature. Transaminases at least twice the upper limit of normal are a severe feature. Pulmonary edema is a severe feature. Persistent headache that does not respond to medication, or cannot be explained by another diagnosis, is a severe feature. Visual symptoms are a severe feature.

Do not let the word "symptom" make you casual. In this disease, symptoms may be organ warnings.

Right upper quadrant pain may be liver capsule tension or HELLP syndrome. Chest pain or dyspnea may be pulmonary edema. Headache and visual symptoms may be cerebral edema, vasospasm, PRES, or impending eclampsia. Nausea and vomiting in the wrong context may be hepatic disease, not simple pregnancy discomfort.

Now assess the fetus and placenta.

At diagnosis, obtain fetal growth assessment when gestational age allows, amniotic fluid assessment, and umbilical artery Doppler if growth restriction or placental insufficiency is suspected. Antenatal testing such as NST or CTG follows gestational age and local protocol.

Guidelines differ in how they label fetal disease. ACOG does not call fetal growth restriction a severe feature by itself. ISSHP and SOMANZ place uteroplacental dysfunction more directly inside the definition of pre-eclampsia. But all agree clinically: a small fetus, abnormal Doppler, oligohydramnios, abruption, or fetal compromise changes risk.

Now where do angiogenic biomarkers fit?

sFlt-1 to PlGF is not another urine test. It is a placental biology test. It asks whether the circulation shows antiangiogenic imbalance.

In PROGNOSIS, a ratio of thirty eight or lower was excellent for short-term rule-out in suspected preterm pre-eclampsia. NICE recommends PlGF based testing to help rule out suspected preterm pre-eclampsia between twenty and thirty six plus six weeks. ISSHP recommends adding angiogenic testing when available for suspected pre-eclampsia before thirty seven weeks. SOMANZ is more cautious about using it as a diagnostic criterion.

The rule is: biomarkers refine risk. They do not rescue a dangerous bedside picture.

Now let us handle the differential diagnosis.

Renal disease is the classic mimic. Chronic kidney disease can cause hypertension and proteinuria before pregnancy. Baseline creatinine, baseline proteinuria, urine sediment, hematuria, casts, and postpartum persistence help separate chronic renal disease from pre-eclampsia. But remember, renal disease and pre-eclampsia can coexist.

Lupus nephritis and antiphospholipid syndrome can mimic pre-eclampsia with hypertension, proteinuria, thrombocytopenia, renal dysfunction, and fetal growth disease. Complement levels, anti-dsDNA, autoimmune history, thrombosis history, and urine sediment may help.

HELLP syndrome overlaps with TTP, HUS, acute fatty liver of pregnancy, viral hepatitis, gallbladder disease, and pancreatitis. If hemolysis or thrombocytopenia worsens after delivery, or the renal and neurologic picture is atypical, do not assume it is ordinary HELLP.

Neurologic symptoms require respect. Headache and seizure can be eclampsia, but they can also be migraine, cerebral venous thrombosis, intracranial hemorrhage, ischemic stroke, epilepsy, tumor, infection, or metabolic disease. The correct stance is

not to ignore pre-eclampsia, and not to ignore neurology. It is to stabilize the mother and investigate when the story does not fit.

Finally, remember pheochromocytoma and secondary hypertension when the blood pressure is extremely labile, refractory, or accompanied by episodic headache, sweating, palpitations, or hypokalemia. It is rare, but missing it can be fatal.

Closing Synthesis

Let us close with the diagnostic sentence.

This is chronic hypertension, GHTN, PET, or superimposed PET, at this gestational age, with or without severe features, with this maternal organ status, and this fetal/placental status.

If the sentence is incomplete, keep evaluating.

Resident Reflex

The resident reflex is this: do not diagnose hypertensive pregnancy from a dipstick and a single cuff reading. Confirm the BP, quantify proteinuria, look for severe features, check maternal organs, assess the fetus and placenta, and use biomarkers only as an adjunct to the whole clinical picture.